

1 Calculating the Phase Shifts

Let us recall that the general form for the radial function of our scattering problem is determined by

$$\frac{d^2 u}{dr^2} = -k^2 u(r) \quad (1)$$

or in terms of the radial function $R_\ell(r) = u_\ell(r)/r$,

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR_\ell}{dr} \right) + \left[k^2 - \frac{\ell(\ell+1)}{r^2} \right] R_\ell(r) = 0. \quad (2)$$

The general form of our solutions at large r can be found by taking the asymptotic forms of the spherical Bessel functions $j_\ell(kr)$ and $n_\ell(kr)$:

$$\begin{aligned} R_\ell(r) &= u_\ell(r)/r = A_\ell j_\ell(kr) + B_\ell n_\ell(kr) \\ &\rightarrow A_\ell \frac{\sin(kr - \ell\pi/2)}{kr} + B_\ell \frac{-\cos(kr - \ell\pi/2)}{kr} \\ &= \frac{1}{2i} \left(\frac{(A_\ell - iB_\ell)e^{i(kr - \ell\pi/2)} - (A_\ell + iB_\ell)e^{-i(kr - \ell\pi/2)}}{kr} \right). \end{aligned} \quad (3)$$

We would like to compare this with the form of the radial wavefunction in our previous notes,

$$R_\ell(r) \rightarrow C_\ell \sin(kr - \ell\pi/2 + \delta_\ell) = C_\ell [\cos \delta_\ell \sin(kr - \ell\pi/2) + \sin \delta_\ell \cos(kr - \ell\pi/2)]. \quad (4)$$

Equating the two gives

$$\begin{aligned} A_\ell &= C_\ell \cos \delta_\ell, \\ B_\ell &= -C_\ell \sin \delta_\ell. \end{aligned} \quad (5)$$

This means that the phase shift δ_ℓ can be determined from the ratio of the coefficients A_ℓ and B_ℓ of the asymptotic wavefunction:

$\tan \delta_\ell = -\frac{B_\ell}{A_\ell}.$

(6)

In practice, to compute the phase shifts, we need to solve the radial Schrödinger equation with the potential $V(r)$, find the solution that is regular at the origin, and then match its asymptotic form to the free solution to extract the approximate solution for A_ℓ and B_ℓ .

Note that in the absence of scattering (no potential), we must have a regular solution, so $A_\ell = 1$ and $B_\ell = 0$, so $\delta_\ell = 0$ as expected. When the potential is turned on, it modifies the wavefunction and introduces a nonzero B_ℓ , which leads to a nonzero phase shift δ_ℓ .

1.1 Hard Sphere Scattering

Let us consider the scattering of a particle off a hard sphere of radius a . By a hard sphere we mean the central potential

$$V(r) = \begin{cases} \infty, & r < a, \\ 0, & r \geq a. \end{cases} \quad (7)$$

The wavefunction must vanish inside the sphere. The boundary condition at $r = a$ is, therefore, $u_\ell(a) = 0$.

Going back to the original form of the solution involving Bessel functions, we write

$$R_\ell(a) = \frac{u_\ell(a)}{a} = A_\ell j_\ell(ka) + B_\ell n_\ell(ka) = 0 \quad \Rightarrow \quad \frac{B_\ell}{A_\ell} = -\frac{j_\ell(ka)}{n_\ell(ka)} = -\tan \delta_\ell. \quad (8)$$

Recall the cross section expansion was just

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2 = \frac{1}{k^2} \left| \sum_{\ell=0}^{\infty} (2\ell+1) e^{i\delta_\ell} \sin \delta_\ell P_\ell(\cos \theta) \right|^2. \quad (9)$$

After angular integration and using the orthonormality of the Legendre polynomials,

$$\sigma = \int d\Omega \frac{d\sigma}{d\Omega} = \frac{4\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell+1) \sin^2 \delta_\ell. \quad (10)$$

By explicit computation, you can check that the ratio of Bessel functions at large radii behaves as

$$\frac{j_\ell(ka)}{n_\ell(ka)} \rightarrow -\tan\left(ka - \frac{\ell\pi}{2}\right). \quad (11)$$

Therefore, at large radii,

$$\delta_\ell \rightarrow -\left(ka - \frac{\ell\pi}{2}\right). \quad (12)$$

Let us investigate the limit $ka \ll 1$. We may call this limit the low-energy or long-wavelength limit, since the de Broglie wavelength $\lambda = 2\pi/k$ of the particle is much larger than the size a of the scatterer. In addition, since the phase shifts for $\ell > 0$ are suppressed by powers of ka , the scattering is dominated by the s-wave. Going back to Eq. (9), we can show that the differential cross section is approximately isotropic,

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2 \simeq \frac{\sin^2 \delta_0}{k^2} + (\ell > 0 \text{ terms}). \quad (13)$$

For $\ell = 0$, with small-angle approximation, $\delta_0 \simeq -ka$,

$$\boxed{\sigma \simeq \frac{4\pi}{k^2} \sin^2(ka) \simeq 4\pi a^2.} \quad (14)$$

This is a very interesting result: the total cross section for scattering off a hard sphere of radius a is approximately the total surface area of the sphere (I say approximately because the higher ℓ terms do not follow this, see below). Naively, you might have expected the cross section to be $\sigma \simeq \pi a^2$, but the wave nature of the “projectile” is clearly important here, especially for $ka \ll 1$.

1.1.1 Higher partial waves: p-wave and d-wave

The general phase shift formula for the hard-sphere scattering case $\delta_\ell = \arctan(j_\ell(ka)/n_\ell(ka))$ can be evaluated for each ℓ using the explicit Bessel functions. In particular, for p -wave scattering ($\ell = 1$): Using $j_1(\rho) = \sin \rho / \rho^2 - \cos \rho / \rho$ and $n_1(\rho) = -\cos \rho / \rho^2 - \sin \rho / \rho$:

$$\tan \delta_1 = -\frac{j_1(ka)}{n_1(ka)} = -\frac{\sin(ka) - ka \cos(ka)}{\cos(ka) + ka \sin(ka)}. \quad (15)$$

At low energies ($ka \ll 1$), we expand $j_1(\rho) \simeq \rho/3$ and $n_1(\rho) \simeq -1/\rho^2$, giving

$$\tan \delta_1 \simeq -\frac{(ka)^3}{3}, \quad (16)$$

which confirms the expected scaling $\delta_\ell \sim (ka)^{2\ell+1}$ for $\ell = 1$. The p-wave contribution to the cross section is therefore

$$\sigma_1 = \frac{12\pi}{k^2} \sin^2 \delta_1 \simeq \frac{12\pi}{k^2} \frac{(ka)^6}{9} = \frac{4\pi a^2}{3} (ka)^4 = \frac{4\pi a^2}{3} \left(\frac{2\pi a}{\lambda}\right)^4, \quad (17)$$

which is suppressed by $(ka)^4$ relative to the s-wave result $\sigma_0 \simeq 4\pi a^2$. Another way to see this suppression is to count powers of velocity $v = k/m$,

$$\sigma_1 \simeq \frac{4\pi a^2}{3} (am)^4 v^4, \quad (18)$$

so the p-wave contribution is suppressed by v^4 relative to the s-wave.

For d -wave scattering, one can show that $\delta_2 \sim (ka)^5$ at low energies, so the d-wave contribution to the cross section is suppressed as $\sigma_2 \propto v^8$.



This velocity suppression at low energies occurs because the centrifugal barrier $\ell(\ell + 1)/(2mr^2)$ prevents higher partial waves from reaching the scattering region. For a hard sphere of radius a at low energies, we can use the approximation to the Bessel functions

$$j_\ell(ka) \simeq \frac{(ka)^\ell}{(2\ell + 1)!!}, \quad n_\ell(ka) \simeq -\frac{(2\ell - 1)!!}{(ka)^{\ell+1}}, \quad (19)$$

at small arguments to show that the phase shifts scale as

$$\delta_\ell \simeq \frac{j_\ell(ka)}{n_\ell(ka)} = -\frac{(ka)^{2\ell+1}}{(2\ell + 1)!!^2 (2\ell - 1)!!}, \quad \sigma_\ell \propto (ka)^{4\ell} \propto v^{4\ell}, \quad (20)$$

where $(2\ell + 1)!! = 1 \times 3 \times 5 \times \dots \times (2\ell + 1)$.

1.1.2 High-energy scattering and the extinction paradox

What happens in the opposite limit, $ka \gg 1$, where the de Broglie wavelength is much smaller than the sphere? In this regime many partial waves contribute, and we can no longer truncate the sum at $\ell = 0$. However, a simple semiclassical argument lets us evaluate the sum approximately.

Take a classical particle with momentum k and impact parameter b (the impact parameter is the perpendicular distance of the particle's trajectory from the center of the sphere, i.e., the distance of closest approach if there were no deflection of the particle's trajectory). Such a trajectory implies an angular momentum $L = |\mathbf{r} \times \mathbf{p}| = kb$. Such an angular momentum mode probes the $\ell = kb$ partial wave, since $\ell(\ell + 1) \simeq (kb)^2$ for large ℓ . Note Partial waves with $\ell \lesssim ka$ correspond to impact parameters $b \lesssim a$: these particles “hit” the sphere and we expect them to be strongly scattered. Partial waves with $\ell \gg ka$ pass far from the sphere and are essentially unaffected: $\delta_\ell \simeq 0$ (recall we are in the short wavelength limit).

For the partial waves that do interact ($\ell \lesssim ka$), the phase shifts δ_ℓ vary rapidly with ℓ and cycle through many multiples of π . To see why, recall from the phase shift formula that $\tan \delta_\ell = j_\ell(ka)/n_\ell(ka)$. When $\ell < ka$, the Bessel functions j_ℓ and n_ℓ are both oscillatory in their argument, so δ_ℓ itself is of order $ka \gg 1$. As ℓ increments by one, δ_ℓ changes by an amount of order unity (comparable to the spacing between zeros of the Bessel functions), so δ_ℓ sweeps through many multiples of π as ℓ runs from 0 to ka . We can therefore replace $\sin^2 \delta_\ell$ by its average value of $1/2$:

$$\sigma \simeq \frac{4\pi}{k^2} \sum_{\ell=0}^{\ell_{\max}} (2\ell + 1) \cdot \frac{1}{2}, \quad (21)$$

where $\ell_{\max} \simeq ka$. Using $\sum_{\ell=0}^N (2\ell + 1) = (N + 1)^2$ (we encountered something similar for m_ℓ in counting the degeneracy of the Hydrogen atom), we get

$$\sigma \simeq \frac{2\pi}{k^2} (ka + 1)^2 \simeq \frac{2\pi}{k^2} (ka)^2 = 2\pi a^2, \quad (22)$$

where the last step holds in the high-energy limit $ka \gg 1$. The scattering is clearly proportional to the geometric shadow of the sphere (area πa^2) but the total cross section is actually twice that, $\sigma \simeq 2\pi a^2$. This fact is sometimes called the “extinction paradox”, because it seems to say that the sphere removes twice as much flux as it geometrically intercepts. Clearly this is a wave phenomenon. You can find intuitive explanations online, though there does seem to be some disagreement on the exact origin of the factor of 2.

Combining the low- and high-energy results, the total hard sphere cross section interpolates between

$$\sigma \simeq \begin{cases} 4\pi a^2, & ka \ll 1 \quad (\text{s-wave dominated}), \\ 2\pi a^2, & ka \gg 1 \quad (\text{semiclassical/diffraction}). \end{cases} \quad (23)$$

Note that the low-energy cross section is *larger* than the high-energy one.

1.2 Scattering off a Spherical Delta Shell

The hard sphere taught us the basic mechanics of partial-wave scattering, but it is too rigid: because the potential is infinitely repulsive, it cannot support bound states, resonances, or any of the richer physics that appears in realistic scattering problems. We now consider a potential that is only slightly more complicated, but physically more interesting. Consider

$$V(r) = V_0 \delta(r - a) \quad (24)$$

which describes a potential concentrated on a sphere of radius a . Here $V_0 > 0$ is repulsive and $V_0 < 0$ is attractive. Despite its simplicity, this potential supports both bound states and scattering resonances, and it reduces to the hard sphere in the limit $V_0 \rightarrow +\infty$. It is therefore the natural next step after the hard sphere.

1.2.1 The s-wave phase shift

The radial Schrödinger equation for $\ell = 0$ reads

$$u''(r) + [k^2 - 2m V_0 \delta(r - a)] u(r) = 0, \quad (25)$$

where $u(r) = r R_0(r)$ as before and the prime means derivative wrt r . Away from $r = a$, this is just the free equation $u'' + k^2 u = 0$. The solutions now must be regular at $r = 0$ and take the asymptotic phase-shift form at $r \rightarrow \infty$:

$$u(r) \sim \begin{cases} \sin(kr), & r < a, \\ C \sin(kr + \delta_0), & r > a. \end{cases} \quad (26)$$

Matching conditions for a delta function potential. Integrating the radial equation from $a - \epsilon$ to $a + \epsilon$ and sending $\epsilon \rightarrow 0$:

$$u'(a^+) - u'(a^-) = 2m V_0 u(a). \quad (27)$$

The wavefunction itself is continuous at $r = a$, since a delta function in u'' produces at most a finite jump in u' . These two conditions are directly analogous to the matching conditions you saw for the one-dimensional delta function potential in QM-I.

Continuity at $r = a$ gives

$$\sin(ka) = C \sin(ka + \delta_0). \quad (28)$$

The derivative condition gives

$$Ck \cos(ka + \delta_0) - k \cos(ka) = 2m V_0 \sin(ka). \quad (29)$$

From that, we get $C = \sin(ka)/\sin(ka + \delta_0)$ and using $\sin(ka)\cos(ka + \delta_0) - \cos(ka)\sin(ka + \delta_0) = -\sin\delta_0$, gives

$$-\frac{k \sin \delta_0}{\sin(ka + \delta_0)} = 2mV_0 \sin(ka). \quad (30)$$

Furthermore, using $\sin(ka + \delta_0) = \sin(ka)\cos\delta_0 + \cos(ka)\sin\delta_0$, we can show that

$$\boxed{\tan \delta_0 = -\frac{\beta \sin^2(ka)}{ka + \beta \sin(ka) \cos(ka)}, \quad \beta \equiv 2mV_0 a.} \quad (31)$$

The dimensionless parameter β controls the strength of the potential: $\beta > 0$ is repulsive, $\beta < 0$ is attractive. First, note that when $V_0 \rightarrow 0$, we recover $\delta_0 \rightarrow 0$ as expected. This single formula contains a lot of physics. Let's study it under some limits. (Recall that we are still in $\ell = 0$ limit. The general case is even richer.)

The hard sphere. Taking $\beta \rightarrow +\infty$ (infinitely repulsive shell):

$$\tan \delta_0 \rightarrow -\frac{\sin^2(ka)}{(ka/\beta) + \sin(ka) \cos(ka)} \rightarrow -\frac{\sin(ka)}{\cos(ka)} = -\tan(ka), \quad (32)$$

so $\delta_0 \rightarrow -ka$, recovering the hard sphere result. This makes sense: an infinitely repulsive shell forces the wavefunction to vanish at $r = a$, just as the hard sphere does.

Weak potential. For $|\beta| \ll 1$:

$$\delta_0 \simeq -\frac{\beta \sin^2(ka)}{ka}, \quad (33)$$

As we will see later, this is the Born approximation result for this potential. The s-wave cross section can be found with a small-angle approximation (because $\beta \ll 1$):

$$\sigma_0 = \frac{4\pi \sin^2 \delta_0}{k^2} \simeq 4\pi \beta^2 a^2 \frac{\sin^4(ka)}{(ka)^4} \xrightarrow{ka \ll 1} 4\pi (\beta a)^2. \quad (34)$$

The cross section is proportional to $(a\beta)^2$, so it is the same for attractive and repulsive potentials of the same strength.

General low energy limit ($ka \ll 1$). Using $\sin(ka) \simeq ka$ and $\cos(ka) \simeq 1$:

$$\delta_0 \simeq -\frac{\beta ka}{1 + \beta}, \quad (35)$$

so the cross section is

$$\sigma_0 \simeq 4\pi a^2 \frac{\beta^2}{(1 + \beta)^2}. \quad (36)$$

This recovers our limit above.

Ramsauer-Townsend transparency. From Eq. (31), $\delta_0 = 0$ whenever $\sin(ka) = 0$, i.e., $ka = n\pi$ for positive integer n . At these special momenta, the s-wave scattering cross section vanishes entirely, regardless of the strength β of the potential. The physical reason is that at $ka = n\pi$, the free wavefunction $\sin(kr)$ already has a node at $r = a$, so $u(a) = 0$ and the delta function has nothing to act on. The particle is blind to the scatterer. This phenomenon is observed experimentally as anomalously low scattering cross sections for electrons on noble gas atoms at specific energies.

2 Resonances and Bound States

So far we have treated the phase shifts δ_ℓ as quantities to be computed from a given potential. In this section we ask: what does the energy dependence of δ_ℓ tell us about the physics? We will also connect this partial-wave language to the S -matrix and to the Breit-Wigner formula derived in the homework.

2.0.1 The S -matrix

As we have argued before, the asymptotic radial wavefunction for the ℓ -th partial wave has the form

$$u_\ell(r) \xrightarrow{r \rightarrow \infty} A_\ell^{\text{in}} e^{-ikr} + A_\ell^{\text{out}} e^{+ikr}, \quad (37)$$

with $A_\ell^{\text{out}} = e^{2i\delta_\ell} A_\ell^{\text{in}}$. We define the (diagonal element of the) S -matrix for partial wave ℓ as

$$S_\ell \equiv e^{2i\delta_\ell} = \frac{A_\ell^{\text{out}}}{A_\ell^{\text{in}}}. \quad (38)$$

Unitarity requires $|S_\ell| = 1$, so S_ℓ lies on the unit circle in the complex plane. The partial-wave scattering amplitude can then be written in a seemingly bizarre way,

$$f_\ell \equiv \frac{e^{i\delta_\ell} \sin \delta_\ell}{k} = \frac{S_\ell - 1}{2ik} = \frac{1}{k} \frac{1}{\cot \delta_\ell - i}. \quad (39)$$

The modulus squared gives the partial cross section, but writing it in this fashion will be useful for understanding resonances and bound states. By the way, we can also derive from the above that the S -matrix is just

$$S_\ell = 1 + 2ikf_\ell, \quad (40)$$

which is the partial-wave version of the relation $S = 1 + iT$, where T is the transition matrix in scattering theory that you may have encountered before.



The Argand diagram. Since $|S_\ell| = 1$, the partial-wave amplitude $f_\ell = (S_\ell - 1)/(2ik)$ traces a circle in the complex plane as δ_ℓ varies from 0 to π . Writing $kf_\ell = e^{i\delta_\ell} \sin \delta_\ell$, we have

$$\Re(kf_\ell) = \frac{1}{2} \sin 2\delta_\ell, \quad \Im(kf_\ell) = \sin^2 \delta_\ell. \quad (41)$$

It is simple to check that these satisfy $(\Re(kf_\ell))^2 + (\Im(kf_\ell) - \frac{1}{2})^2 = \frac{1}{4}$. This is just a circle of radius $1/2$ centered at $(0, 1/2)$ in the complex plane (see Sakurai's discussion for a nice picture). This is called the **unitarity circle**.

There are three special points: $\delta_\ell = 0$ is the origin and corresponds to no scattering; $\delta_\ell = \pi/2$ is the top of the circle, $kf_\ell = i$ and corresponds to maximum scattering ($\sigma_\ell = (4\pi/k^2)(2\ell+1)$, the unitarity bound), and, finally, $\delta_\ell = \pi$ maps back to the origin. As the energy increases, the point traces the circle counterclockwise. As we will see, a resonance corresponds to a rapid traversal through the top of the circle.

2.1 Resonances

A scattering resonance is a quasi-bound state in the continuum. It corresponds to a configuration in which the particle is temporarily trapped by the potential (for example, behind a centrifugal or potential barrier) before escaping. The hallmark of a resonance is a rapid increase of δ_ℓ through $\pi/2$ as the energy sweeps through the resonance energy E_R .

To parameterize this behavior, we expand $\cot \delta_\ell(E)$ near the resonance. Since $\delta_\ell = \pi/2$ at $E = E_R$, we have $\cot \delta_\ell(E_R) = 0$. We then Taylor expand $\cot \delta_\ell(E)$ around $E = E_R$, which is to say, we expand $\cot \delta_\ell$ around $\pi/2$:

$$\cot \delta_\ell(E) = \cot(\pi/2) + \left. \frac{d \cot \delta_\ell}{dE} \right|_{E=E_R} (E - E_R) + \dots \simeq -\frac{2}{\Gamma} (E - E_R), \quad (42)$$

where $\Gamma > 0$ is the width of the resonance, defined by this expansion,

$$\Gamma \equiv -2 \left(\left. \frac{d \cot \delta_\ell}{dE} \right|_{E=E_R} \right)^{-1}. \quad (43)$$

This formula for Γ is a bit abstract and not very useful. We will gain more intuition on Γ in what follows. The sign convention ensures that δ_ℓ increases through $\pi/2$ as E increases through E_R .

Substituting into Eq. (39):

$$f_\ell^{\text{res}} = \frac{1}{k} \frac{\Gamma/2}{(E_R - E) - i\Gamma/2}. \quad (44)$$

The partial cross section is

$$\sigma_\ell = \frac{4\pi(2\ell+1)}{k^2} \sin^2 \delta_\ell = \frac{4\pi(2\ell+1)}{k^2} \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}. \quad (45)$$

This is a Lorentzian peaked at $E = E_R$ with full width at half maximum (FWHM) equal to Γ . At resonance ($E = E_R$), $\sin^2 \delta_\ell = 1$ and the partial cross section saturates the unitarity bound $\sigma_\ell^{\text{max}} = 4\pi(2\ell+1)/k_R^2$.



Compare Eq. (45) for $\ell = 0$ (s-wave, single elastic channel) with the Breit-Wigner formula from HW 8:

$$\sigma = \frac{\pi}{k^2} \frac{\Gamma_a \Gamma_b}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (46)$$

For elastic scattering through a single channel, $\Gamma_a = \Gamma_b = \Gamma$ and the numerator becomes Γ^2 , giving $\sigma = (4\pi/k^2) \Gamma^2 / (4[(E - \tilde{E}_R)^2 + \Gamma^2/4])$. This is exactly Eq. (45) with $\ell = 0$. The homework derived this from the Dyson series (time-dependent perturbation theory), while here we derived it from phase shifts (time-independent scattering theory).

For the S -matrix, the Breit-Wigner parameterization gives

$$S_\ell = e^{2i\delta_\ell} = \frac{\cot \delta_\ell + i}{\cot \delta_\ell - i} = \frac{E_R - E + i\Gamma/2}{E_R - E - i\Gamma/2}. \quad (47)$$

This expression has a *pole* at

$$E = E_R - i\Gamma/2. \quad (48)$$

The pole sits in the lower half of the complex energy plane. We saw this same complex energy $\mathcal{E}_R = E_R - i\Gamma/2$ in the homework, where it arose from the second-order Dyson series. Here it emerges from the analytic structure of the S -matrix. The imaginary part $-\Gamma/2$ corresponds to the exponential decay of the metastable state, $|\psi(t)|^2 \propto e^{-\Gamma t}$ which has a lifetime of $\tau = 1/\Gamma$.